Diego Henriquez

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Education

Mississippi State University, Starkville, MS — Expected Dec 2027
B.S. Industrial & Systems Engineering | Focus Area: Data Analytics — GPA: 3.90/4.00
Honors: President's Scholar (Fall 2025, Spring 2026)
Relevant Coursework: Engineering Statistics I & II (regression, ANOVA, DOE), Production Control, Systems Engineering Analysis, Engineering Economy, Information Systems, Lean Work Systems, Manufacturing Processes

Embry-Riddle Aeronautical University, Daytona Beach, FL — 2019 – 2021
Completed coursework toward B.S. in Aerospace Engineering

Skills

Data Analytics Tools: Excel (PivotTables, PivotCharts, data cleaning), Power BI, Minitab, MATLAB, PowerPoint
Databases: SQL, Microsoft Access
Engineering & Manufacturing: Fusion 360 (CAD/CAM), CNC router setup & operation, G-code generation, 3D printing
Hardware & Prototyping: Arduino, ESP32, sensor integration, stepper motor control
Research: Systematic literature review, synthesis, technical writing
Languages: English, Spanish

Projects

Course Project: Production Throughput & Bottleneck Analysis (Information Systems, Power BI)

- Cleaned 5 raw production, labor, downtime, shift, and workstation datasets and built a Power BI data model, DAX measures, and an interactive dashboard filterable by workstation and shift.
- Identified the Drilling workstation as the system bottleneck (83.9% of unit target vs. 95–104% elsewhere; 75.4% OEE), ruling out scrap and tracing the root cause to machine-fault downtime at 2.6x any other station.
- Wrote an executive summary for plant leadership recommending preventive maintenance, a repair-or-replace equipment inspection, and improved material availability.

Personal Project: Housing Market Trends Dashboard (Power BI)

- Cleaned and prepared 6 Zillow housing datasets in Excel and built a star-schema data model linking them through shared date and metro dimension tables.
- Designed an interactive dashboard with KPI cards, a state map, and state, metro, price-tier, and date filters to compare housing trends from 2013 to 2026.

Course Project: LLM Accuracy & Response Time Study (Engineering Statistics I, Excel)

- Built and analyzed a 400-prompt dataset across 40 topics, scoring accuracy and response time with a standardized key to identify category-level trends and summarize findings.

Undergraduate Research (IE 4800): Systematic Literature Review (AR & Human Performance)

- Screened studies against predefined inclusion and exclusion criteria and synthesized findings across performance metrics (time, accuracy, error rate, workload) to identify trends and research gaps.

Experience

Owner, Emblem Premier LLC – small online business — 2023 – 2026

- Designed sports-themed wall art in Fusion 360 and produced it on a self-built CNC router, handling setup, calibration, troubleshooting, CAM toolpaths, and G-code.
- Sold directly to customers through the company website, managing production, packaging, and shipping.

Activities

Formula SAE Team – Member, Embry-Riddle Aeronautical University — 2020 – 2021

- Supported design reviews by checking part fit/clearances and assisting with rapid prototyping via 3D printing.

Certifications

Google Advanced Data Analytics Professional Certificate – Coursera — Expected Spring 2027
In progress: Python, regression, machine learning, Tableau
Certificate in Data Analysis – Center for Professional Studies — Issued Sep 2026